ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Practical Applications in Complex Materials Systems

I. Review of Kubo Greenwood Formula

II. Optical Conductivity – Probing Light-Matter Interaction

III. Transport Anisotropy Induced by Strain or Defects in Graphene

IV. Kubo-Greenwood in Disordered and Strongly Correlated Systems

I. Review of Kubo Greenwood Formula

Kubo Formula $\sigma_{\mu\nu}(\omega) = \frac{1}{V}\int_0^\infty dt e^{i\omega t}\int_0^\beta d\lambda \langle [J_\mu(t), J_\nu(-i\hbar\lambda)]\rangle_0$

Kubo-Greenwood Formula $\sigma_{\mu\nu}(\omega) = \frac{\pi e^2\hbar}{V}\int dE\left(-\frac{df}{dE}\right) Tr\{v_\mu[G_n^R(E) - G_n^A(E)]v_\nu[G_m^R(E) - G_m^A(E)]\}$

Kubo Formula shows the conductivity as a correlation function between current operators at different times, or equivalently, in the energy domain, as a trace involving velocity operators and Green's functions

The broadening parameter η, which we introduced last time, is not merely a numerical trick – it physically represents the finite lifetime of electronic states due to scattering from impurities, phonons, or other electrons

In this lecture, the focus is on how choosing the right η, combined with accurate Hamiltonians from DFT or Wannier functions, enables the prediction of realistic transport behavior in materials far more complex than ideal crystals

I. Review of Kubo Greenwood Formula

In real materials research, especially in the Chip science, one needs to connect fundamental quantum mechanics with device-level performance, so some practical examples are discussed in this lecture

For instance, in SiC power devices, even trace amounts of metal impurities like Au can significantly affect long-term reliability. In optoelectronic materials, we need accurate optical conductivity to design efficient photodetectors

The Kubo-Greenwood method, when properly implemented with Wannier Hamiltonians or tight-binding models, provides a powerful tool to predict these properties before expensive experiments

♦ Theory alone is not enough for materials design

♦ Bridge microscopic quantum dynamics and macroscopic device performance

♦ Applications in power electronics, optoelectronics, and quantum materials

♦ Challenges: disorder, doping, anisotropy, SOC effects

I. Review of Kubo Greenwood Formula

Case 1: Optical Conductivity in 2D Materials

Case 2: Transport Anisotropy in Doped SiC (Au migration)

Case 3: Disorder and Localization Effects

First, optical conductivity — crucial for understanding light-matter interaction in 2D semiconductors and topological materials

Second, transport anisotropy in Au-doped SiC — directly relevant to the research on impurity migration and power device reliability

Third, how disorder and localization manifest in the Kubo-Greenwood framework — a topic essential for realistic materials containing defects

These examples will prepare us well for the next lecture, where we will dive into the actual code implementation using pristine graphene as a clean test case

I. Review of Kubo Greenwood Formula

♦ **Understand how KG formula is applied beyond ideal systems**

♦ **Learn to interpret computational results in the context of real materials**

♦ **Prepare for numerical implementation in the next session**

II. Optical Conductivity – Probing Light-Matter Interaction

While the DC conductivity discussed in previous lectures tells us about steady-state electron transport, optical conductivity reveals how materials respond to electromagnetic waves at finite frequencies. This is extremely important for optoelectronic devices, photodetectors, and solar cells

The Kubo-Greenwood formula can be naturally extended to finite frequency ω, allowing one to compute the frequency-dependent conductivity $\sigma(\omega)$

$$\sigma_{\mu\nu}(\omega) = \frac{ie^2}{\omega}\int dE f(E) Tr\{v_\mu G^R(E) v_\nu G^A(E+\hbar\omega)\}$$

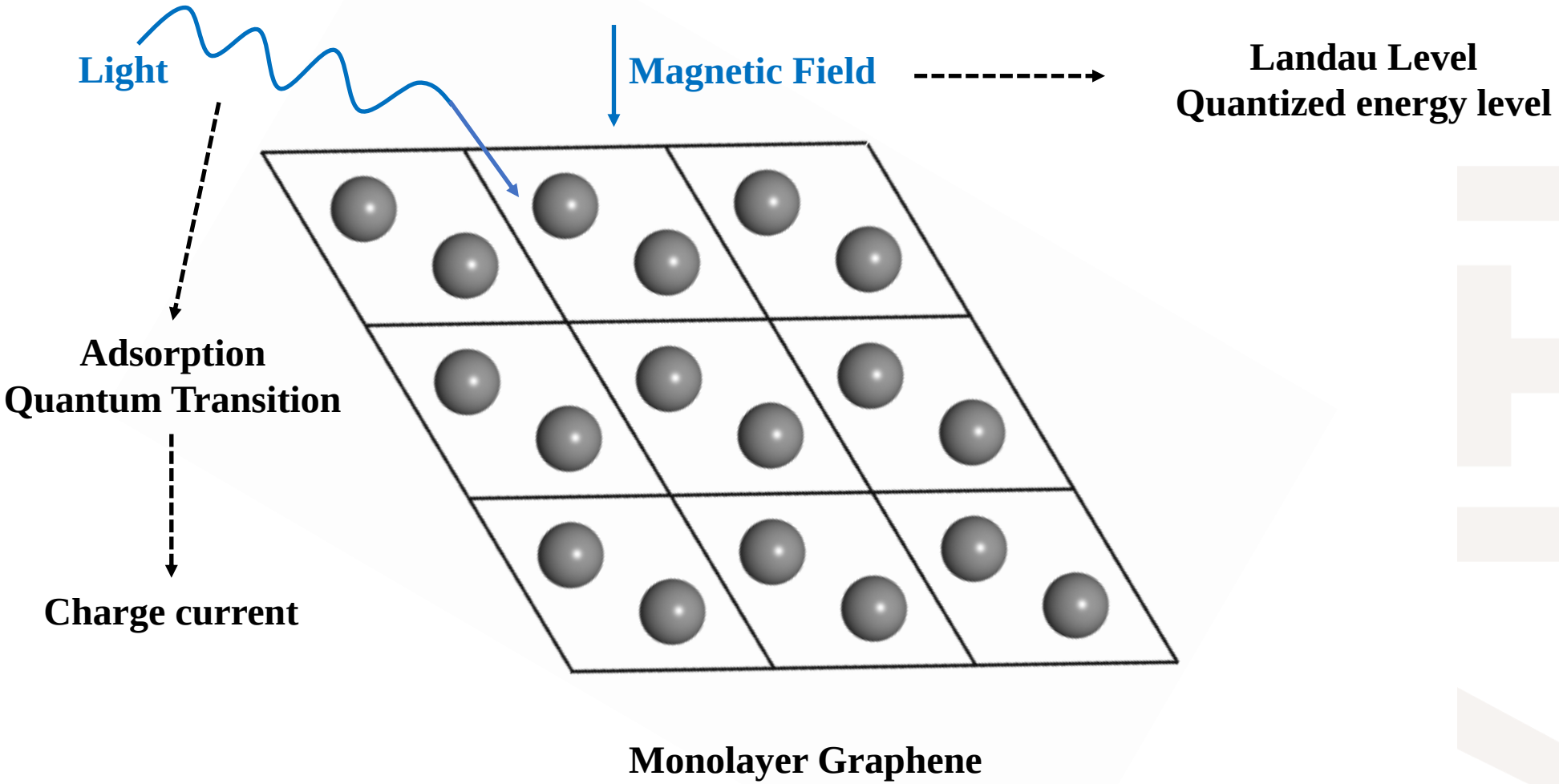
In the frequency domain, the Kubo-Greenwood formula takes the form shown here. Note the appearance of both retarded (G^R) and advanced (G^A) Green functions

The real part of $\sigma(\omega)$ is related to absorption (energy dissipation), while the imaginary part is related to dispersion. Thanks to the Kramers-Kronig relations, they are not independent

In practice, one often focus on Re[$\sigma(\omega)$], which can be directly compared with experimental optical absorption spectra

II. Optical Conductivity – Probing Light-Matter Interaction

Magneto-optical conductivity in graphene



V. PGusynin, S. G. Sharapov and J. P. Carbotte, Magneto-optical conductivity in graphene, *J. Phys.: Condens. Matter* 19 (2007) 026222

II. Optical Conductivity – Probing Light-Matter Interaction

Magneto-optical conductivity in graphene

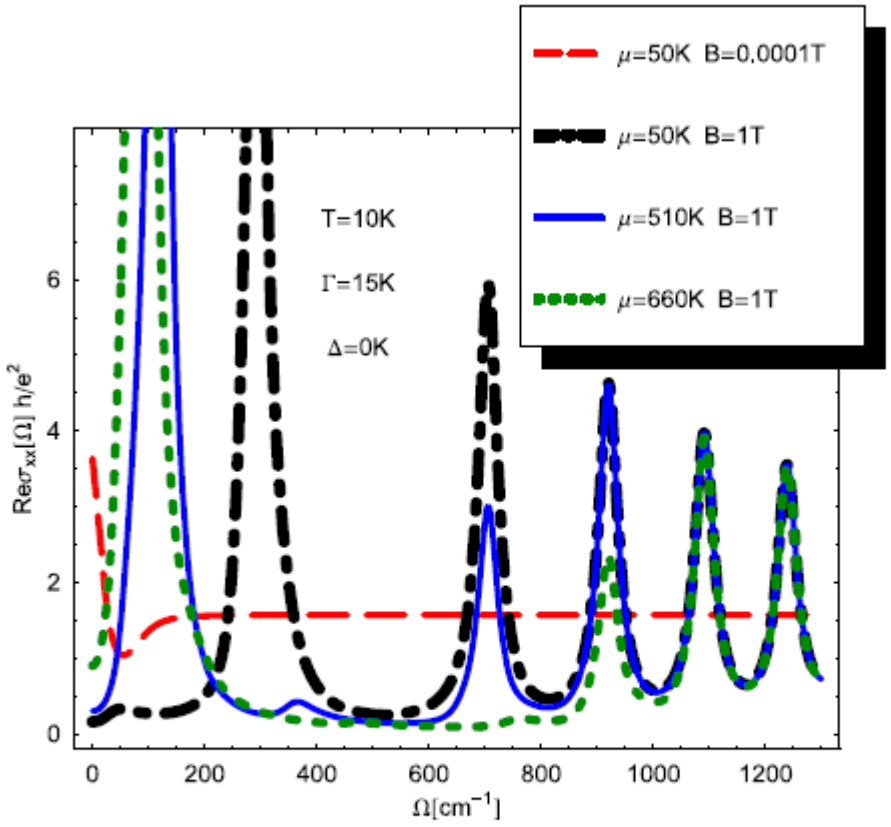


Figure 1. The real part of the longitudinal conductivity, Re $\sigma_{xx}(\Omega)$, in units of e^2/h versus frequency Ω in cm^{-1} for temperature $T = 10$ K and scattering rate $\Gamma = 15$ K. Long dashed, the chemical potential $\mu = 50$ K and the magnetic field $B = 10^{-4}$ T; dash–dotted, $\mu = 50$ K and $B = 1$ T; solid, $\mu = 510$ K and $B = 1$ T; short dashed, $\mu = 660$ K and $B = 1$ T.

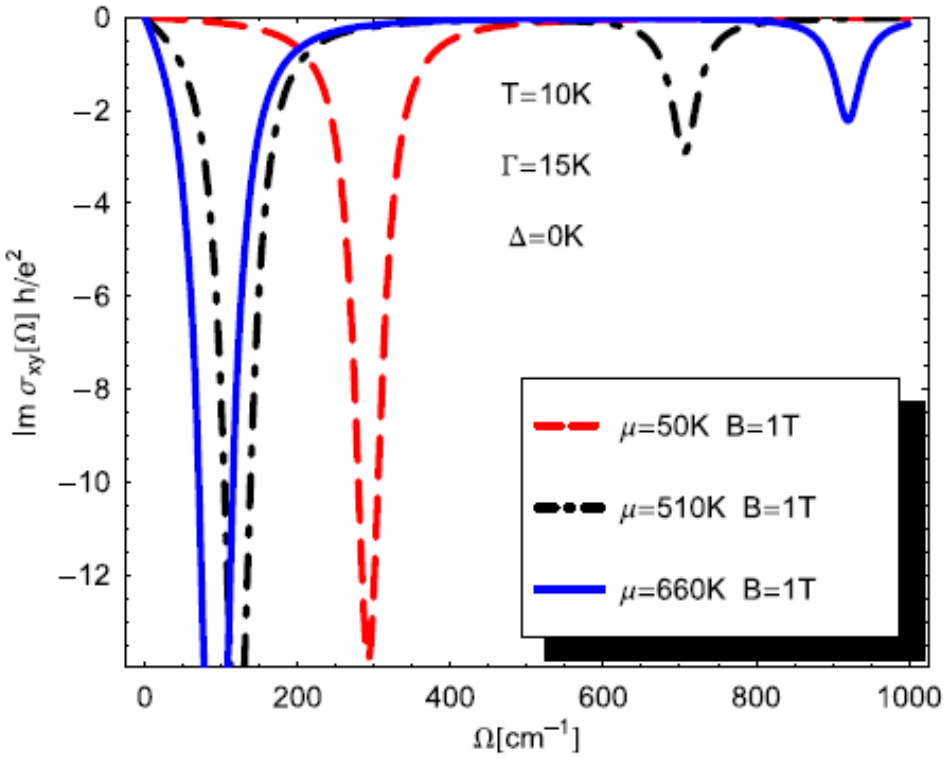


Figure 8. The imaginary part of the Hall conductivity, Im $\sigma_{xy}(\Omega)$ in units of e^2/h as a function of frequency Ω in cm^{-1}. The three cases are for $\mu = 50$ K (long dashed), $\mu = 510$ K (dash–dotted) and $\mu = 660$ K (solid). The other parameters are $B = 1$ T, $T = 10$ K and $\Gamma = 15$ K.

V. PGusynin, S. G. Sharapov and J. P. Carbotte, Magneto-optical conductivity in graphene, *J. Phys.: Condens. Matter* 19 (2007) 026222

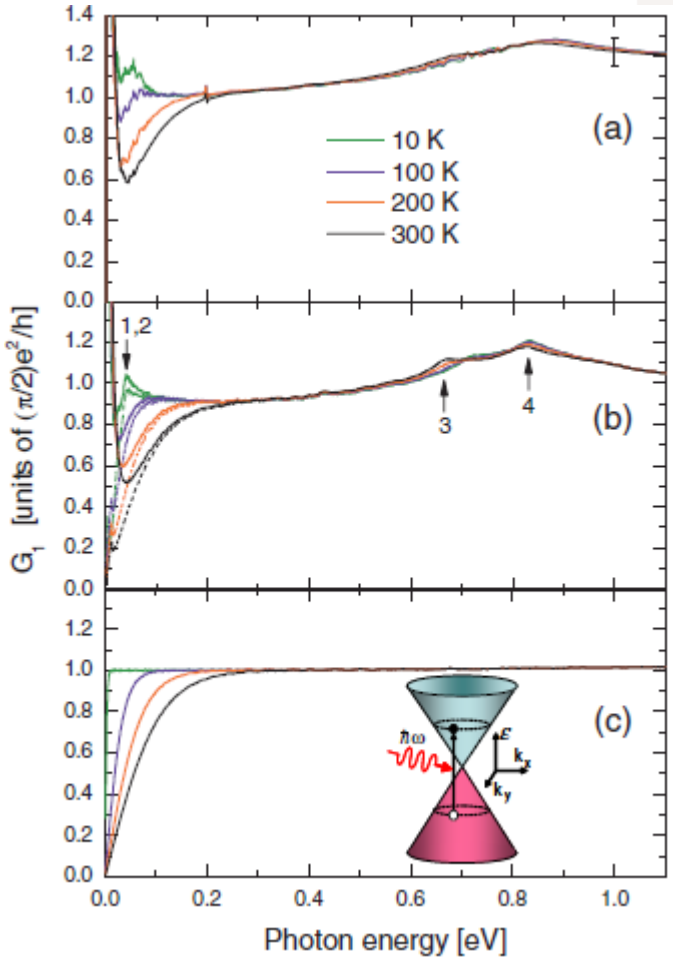
II. Optical Conductivity – Probing Light-Matter Interaction

Optical Conductance of Graphite

$$G_1(\omega) = \frac{\pi e^2}{\omega} |\mathbf{v}(\omega)|^2 D(\omega) \left[f\left(-\frac{\hbar\omega}{2}\right) - f\left(\frac{\hbar\omega}{2}\right) \right]$$

Velocity matrix element **2D joint density of states**

FIG. 1 (color online). The real part of the optical sheet conductance of graphite per layer [(a) experiment, (b) calculation)] as well as the calculated conductance of isolated undoped graphene (c). The error bar indicates the inaccuracy of the absolute value due to systematic experimental uncertainties; the relative inaccuracy of the temperature dependence is much smaller. In (b) the dashed lines indicate the interband conductance only, while the solid line represents the total conductance with a Drude peak added (the scattering rate is 5 meV). In (c) the thermally activated Drude peak is not shown. The arrows in (b) correspond to optical transitions indicated in Fig. 3. A certain noise in (b) and (c) is of numerical origin. The inset of (c) depicts the optical transitions between hole and electron bands in monolayer graphene.

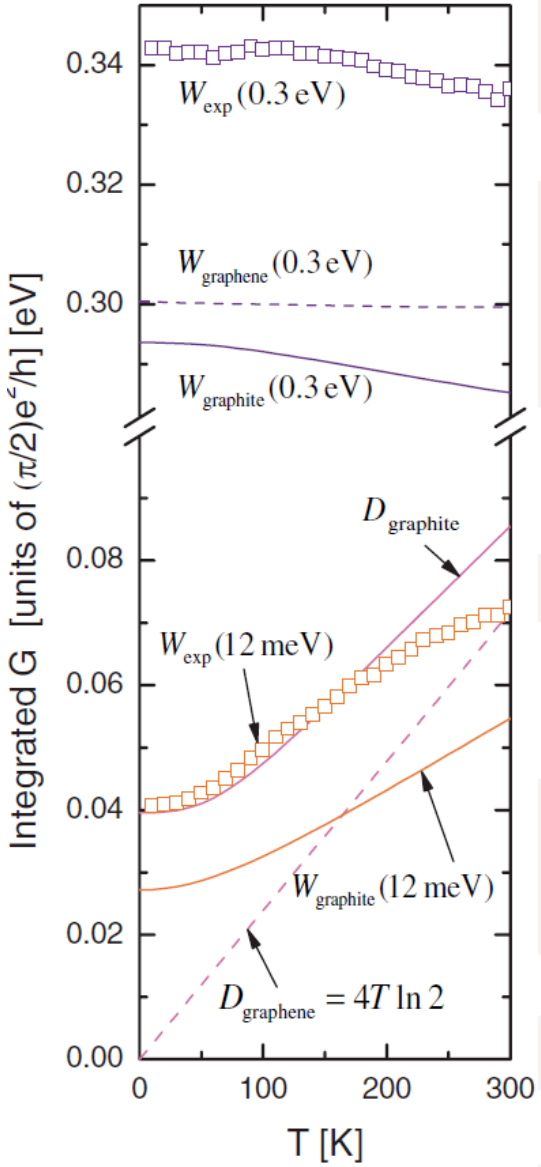


A. B. Kuzmenko, E. van Heumen, F. Carbone and D. van der Marel, Universal Optical Conductance of Graphite, *Phys. Rev. Lett.*, 100, 117401 (2008)

II. Optical Conductivity – Probing Light-Matter Interaction

Optical Conductance of Graphite

FIG. 4 (color online). The integrated conductance of graphite (symbols, experiment; solid lines, calculation) and of monolayer undoped graphene (dashed lines, calculation).



A. B. Kuzmenko, E. van Heumen, F. Carbone and D. van der Marel, Universal Optical Conductance of Graphite, *Phys. Rev. Lett.*, 100, 117401 (2008)

II. Optical Conductivity – Probing Light-Matter Interaction

Magneto-optical conductivity in a topological insulator

$$H_0 = \frac{\hbar^2 k^2}{2m} + \hbar v_F (k_x \sigma_y - k_y \sigma_x)$$

Kinetic energy term

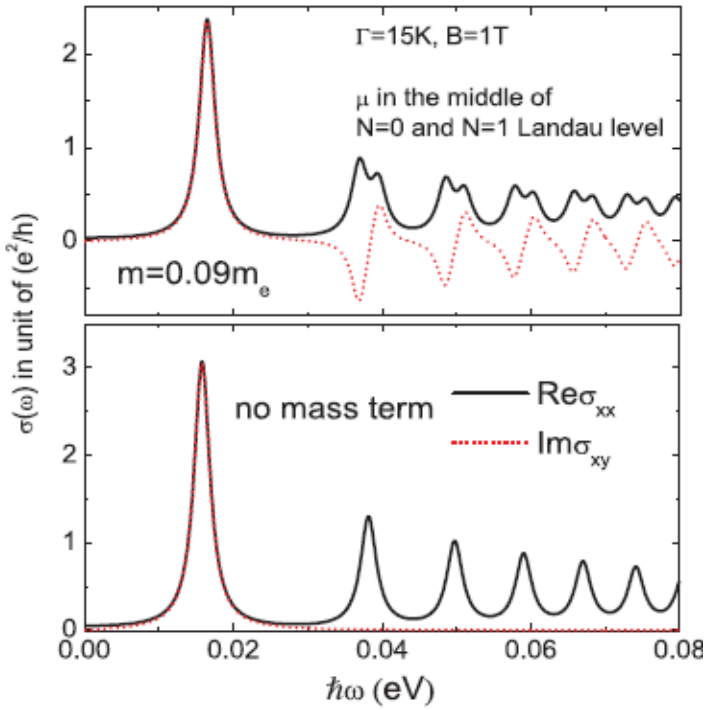
Relativistic Dirac fermions with velocity v_F

Kubo Formula $$\sigma_{\alpha\beta}(\omega) = \frac{-i}{2\pi l_B^2} \sum_{N,N',s,s'} \frac{f_{N,s} - f_{N',s'}}{E_{N,s} - E_{N',s'}} \frac{\langle N,s|j_\alpha|N',s'\rangle \langle N',s'|j_\beta|N,s\rangle}{\omega - E_{N,s} + E_{N',s'} + i/(2\tau)}$$

Zhou Li and J. P. Carbotte, Magneto-optical conductivity in a topological insulator, Phys. Rev. B, 88, 045414 (2013)

II. Optical Conductivity – Probing Light-Matter Interaction

Magneto-optical conductivity in a topological insulator



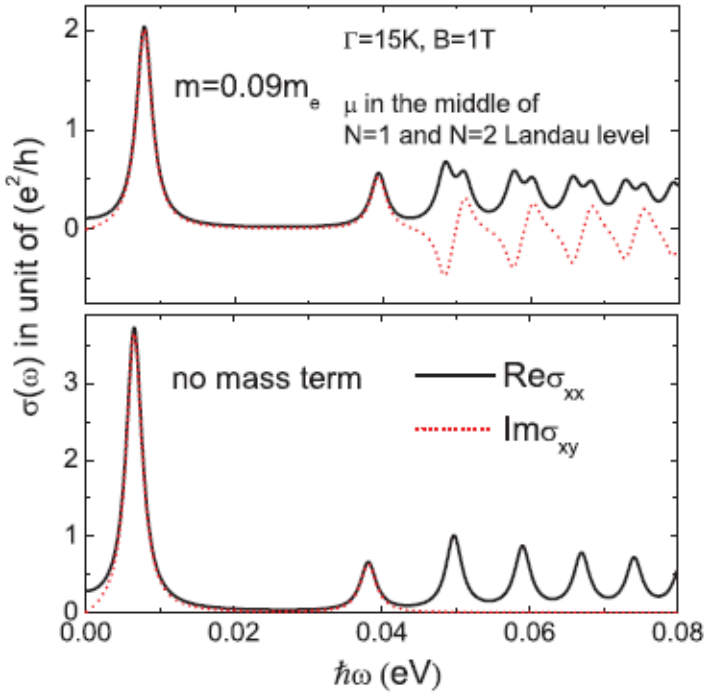


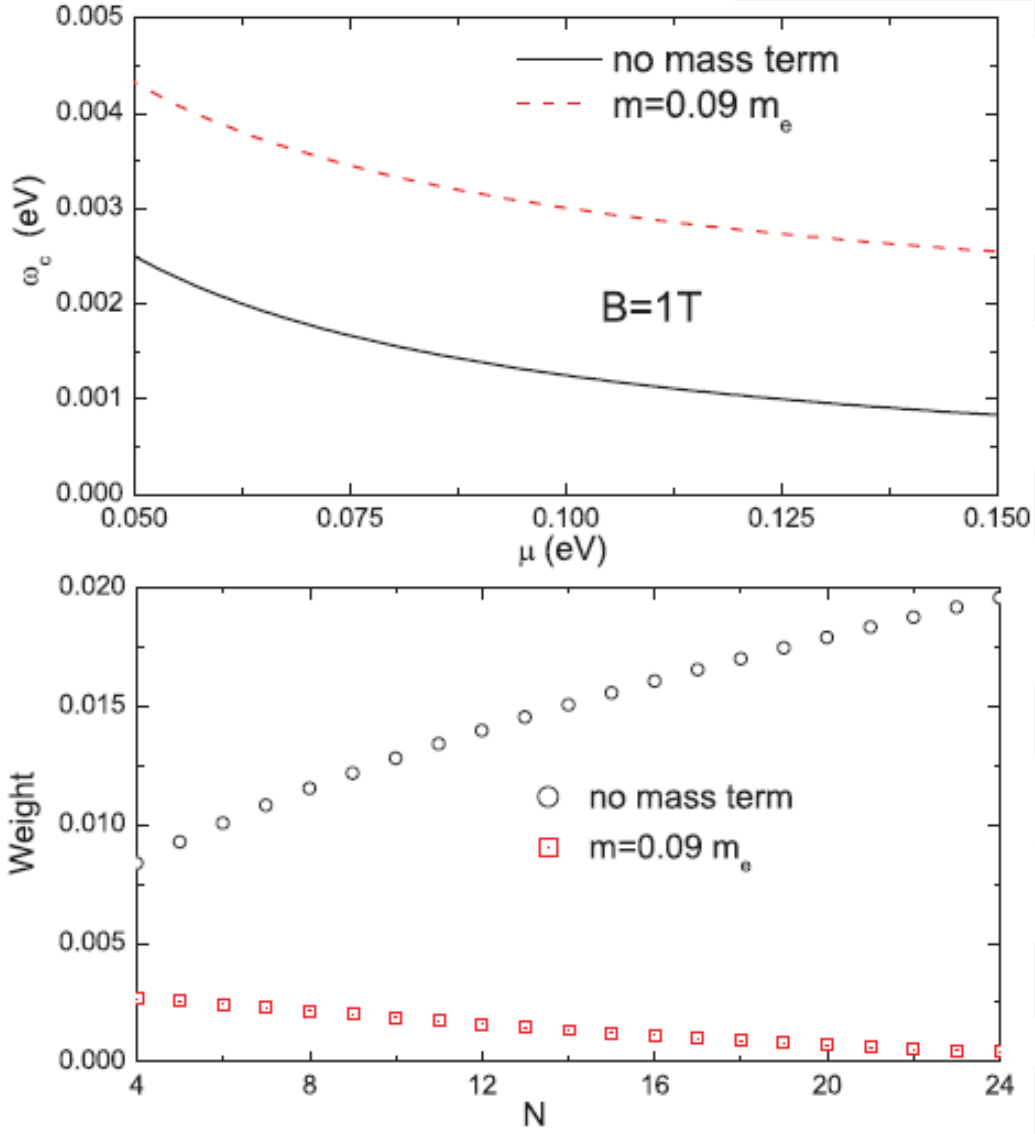
FIG. 4. (Color online) The real part of the longitudinal optical conductivity $\sigma_{xx}(\omega)$ (solid black curve) compared with the imaginary part of the transverse Hall conductivity $\sigma_{xy}(\omega)$ (dotted red curve) in units of e^2/h as a function of photon energy $\hbar\omega$ in eV. The electron mass term has $m = 0.09\ m_e$, the residual scattering rate is $\Gamma = 15\ K$, and the magnetic field $B = 1\ T$. The left two frames are for chemical potential falling between $N = 0$ and $N = 1$ Landau levels while in the right two frames it falls between $N = 1$ and $N = 2$. In each case we show first results for finite m and below for $m = \infty$ (pure Dirac) which is included for comparison. Here $\alpha/\hbar = 4.3\times10^5\ m/s$

Zhou Li and J. P. Carbotte, Magneto-optical conductivity in a topological insulator, Phys. Rev. B, 88, 045414 (2013)

II. Optical Conductivity – Probing Light-Matter Interaction

Magneto-optical conductivity in a topological insulator

FIG. 8. (Color online) The top panel gives the semiclassical cyclotron frequency as a function of chemical potential μ and the bottom panel gives the optical spectral weight under the Nth line for $m = 0.09m_e$ (open square) compared with its value for pure Dirac (open circle). Here $\alpha/\hbar = 4.3 \times 10^5$ m/s.



Zhou Li and J. P. Carbotte, Magneto-optical conductivity in a topological insulator, Phys. Rev. B, 88, 045414 (2013)

III. Transport Anisotropy Induced by Strain or Defects in Graphene

Another application example of Kubo formula is the DC transport anisotropy in graphene — a system that is isotropic in its pristine form but becomes anisotropic under realistic perturbations such as strain or defects. This case study uses the Kubo-Greenwood formula directly and connects beautifully to our code implementation in the next lecture

In ideal graphene, the conductivity is isotropic due to its hexagonal symmetry. However, in real devices, uniaxial strain or defects are almost inevitable. The key question is whether these small structural changes can induce measurable transport anisotropy, and whether the Kubo-Greenwood formula can capture it

- **Kubo-Greenwood formula for σ_{xx} ($\omega = 0$) and σ_{yy} ($\omega = 0$)**

- **k point convergence and broadening η control**

III. Transport Anisotropy Induced by Strain or Defects in Graphene

Strain effect on the optical conductivity of graphene

$$\sigma_{lm}(\mu,T;\omega) = -\frac{1}{\hbar\omega N A_{cell}}\mathrm{Im}\,\tilde{\Pi}^{\mathrm{R}}_{lm}(0,0,\omega), \qquad (26)$$

where $\tilde{\Pi}^{\mathrm{R}}_{lm}$ is the retarded version of

$$\tilde{\Pi}_{lm}(\mathbf{p},\mathbf{p}',\tau) = -\langle T_\tau[\tilde{J}^{\nabla}_l(\mathbf{p},\tau)\tilde{J}^{\nabla}_m(\mathbf{p}',0)]\rangle, \qquad (27)$$

$$\sigma_{lm}(\omega) = \mathrm{Re}\frac{2i}{A_{cell}\hbar\omega}\frac{1}{N}\sum_{\mathbf{k}\lambda} \times\left\{[\tilde{J}^{\nabla}_l(\mathbf{k})]_{\lambda\bar{\lambda}}[\tilde{J}^{\nabla}_m(\mathbf{k})]_{\bar{\lambda}\lambda}\frac{n_F(\xi_{\mathbf{k}\bar{\lambda}}) - n_F(\xi_{\mathbf{k}\lambda})}{\hbar\omega + \xi_{\mathbf{k}\lambda} - \xi_{\mathbf{k}\bar{\lambda}} + i0^+}\right\}, \qquad (28)$$

Kubo Formula

F. M. D. Pellegrino, G. G. N. Angilella and R. Pucci, Strain effect on the optical conductivity of graphene, *Phys. Rev. B* 81, 035411 (2010)

III. Transport Anisotropy Induced by Strain or Defects in Graphene

Strain effect on the optical conductivity of graphene

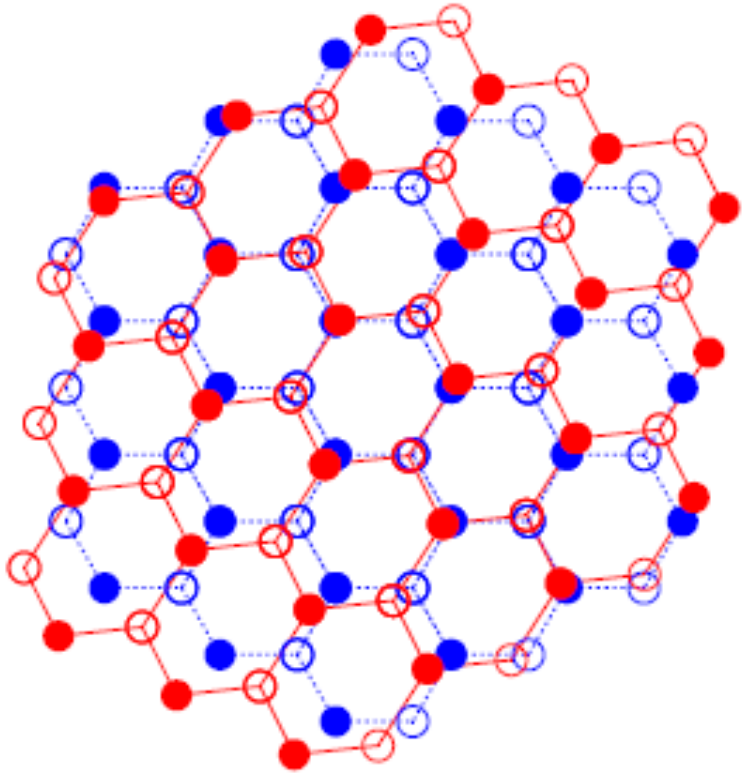


FIG. 1. (Color online) Schematic representation of the strained (red) vs unstrained (blue) honeycomb lattice for $\theta=\pi/4$. Open (closed) symbols refer to the A (B) sublattices, respectively.

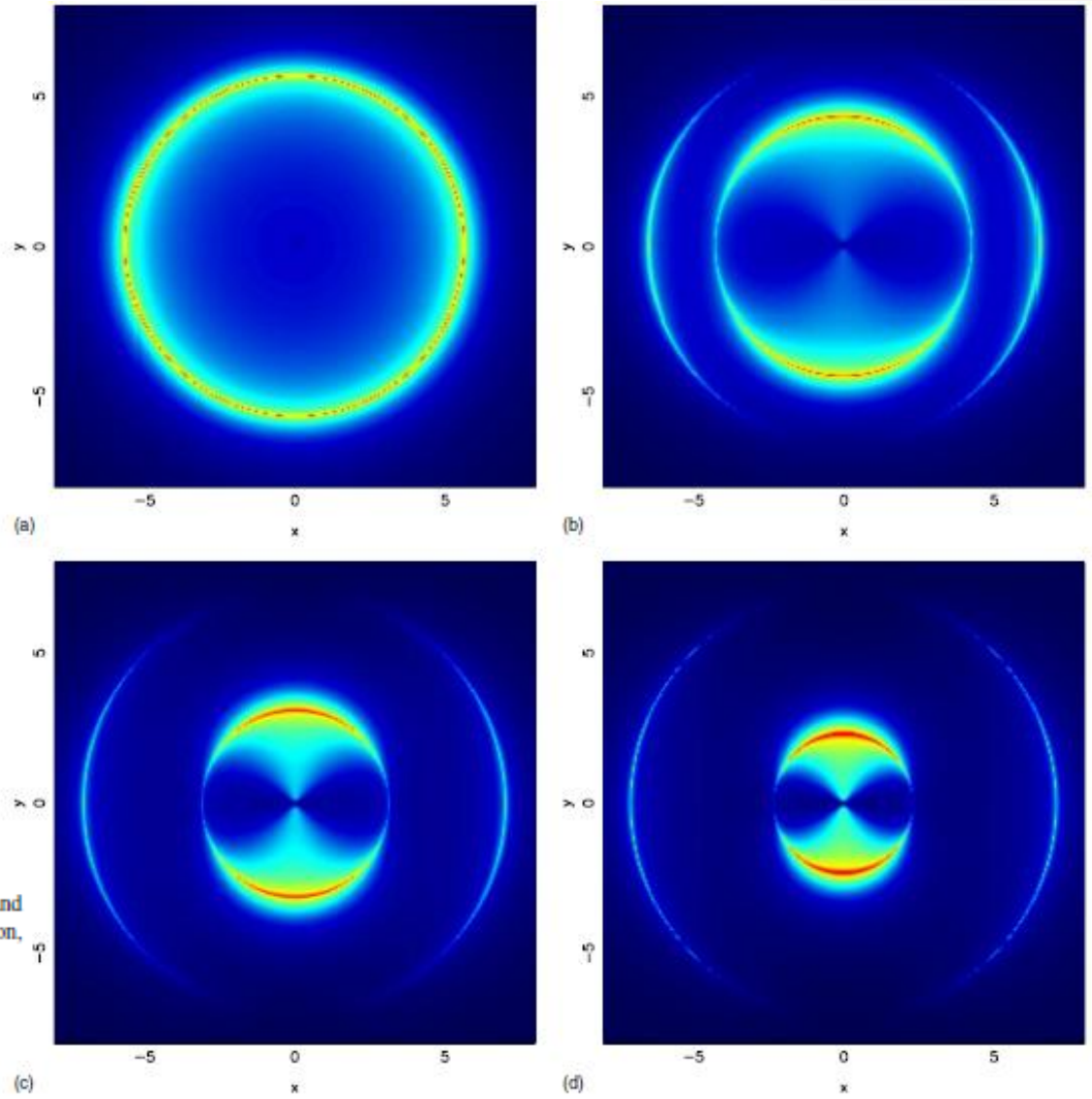


FIG. 7. (Color online) Polar plots of longitudinal-optical conductivity $\sigma_{\parallel}/\sigma_0$, Eq. (29), as a function of frequency $\omega>0$ (polar axis) and electric field orientation ϕ (azimuthal direction). Here, we set $\mu=0$ and $k_BT=0.025$ eV. Strain is applied along the $\theta=0$ (armchair) direction, and the strain modulus increases from left to right, and from top to bottom ($\varepsilon=0, 0.075, 0.175, 0.275$).

F. M. D. Pellegrino, G. G. N. Angilella and R. Pucci, Strain effect on the optical conductivity of graphene, *Phys. Rev. B* 81, 035411 (2010)

III. Transport Anisotropy Induced by Strain or Defects in Graphene

Strain effect on the optical conductivity of graphene

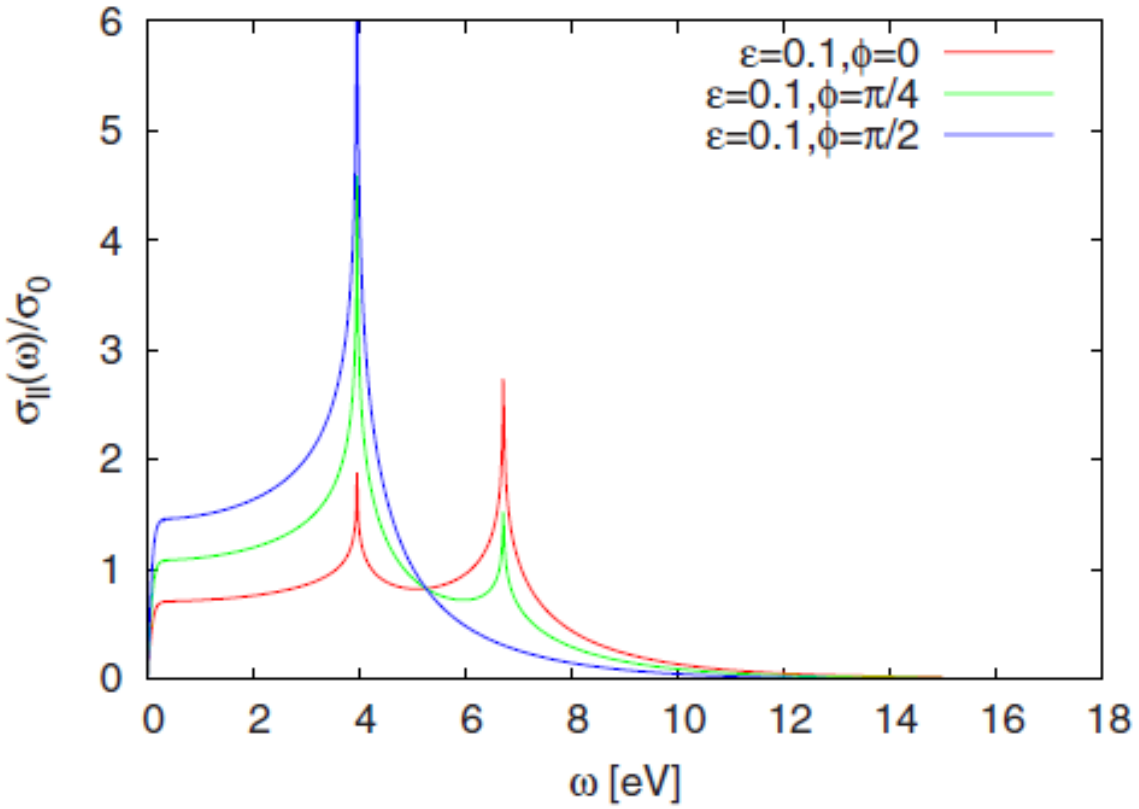


FIG. 8. (Color online) Longitudinal-optical conductivity σ_{ll}/σ_0, Eq. (29), as a function of frequency $\omega>0$, for fixed strain modulus $\varepsilon=0.1$ and strain direction $\theta=0$ (armchair). Different lines refer to various orientations of the electric field ($\phi=0, \pi/4, \pi/2$).

F. M. D. Pellegrino, G. G. N. Angilella and R. Pucci, Strain effect on the optical conductivity of graphene, *Phys. Rev. B* 81, 035411 (2010)

III. Transport Anisotropy Induced by Strain or Defects in Graphene

Transport properties of graphene containing structural defects

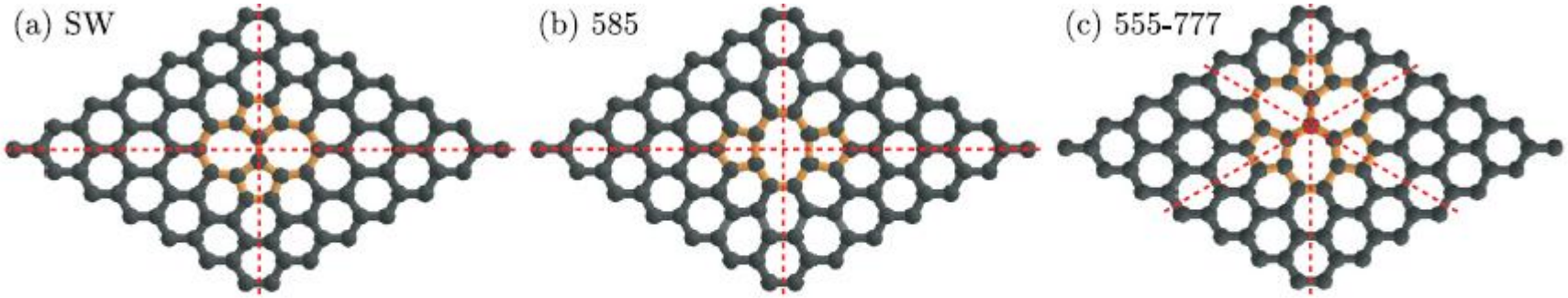


FIG. 1. (Color online) Schematic of the three structural defects: (a) Stone-Wales, (b) 585, and (c) 555-777 divacancies. Symmetry axes are drawn in red dashed lines.

Aurélien Lherbier, Simon M.-M. Dubois, Xavier Declerck, *et al.*, Transport properties of graphene containing structural defects, *Phys. Rev. B* 86 075402 (2012)

III. Transport Anisotropy Induced by Strain or Defects in Graphene

Transport properties of graphene containing structural defects

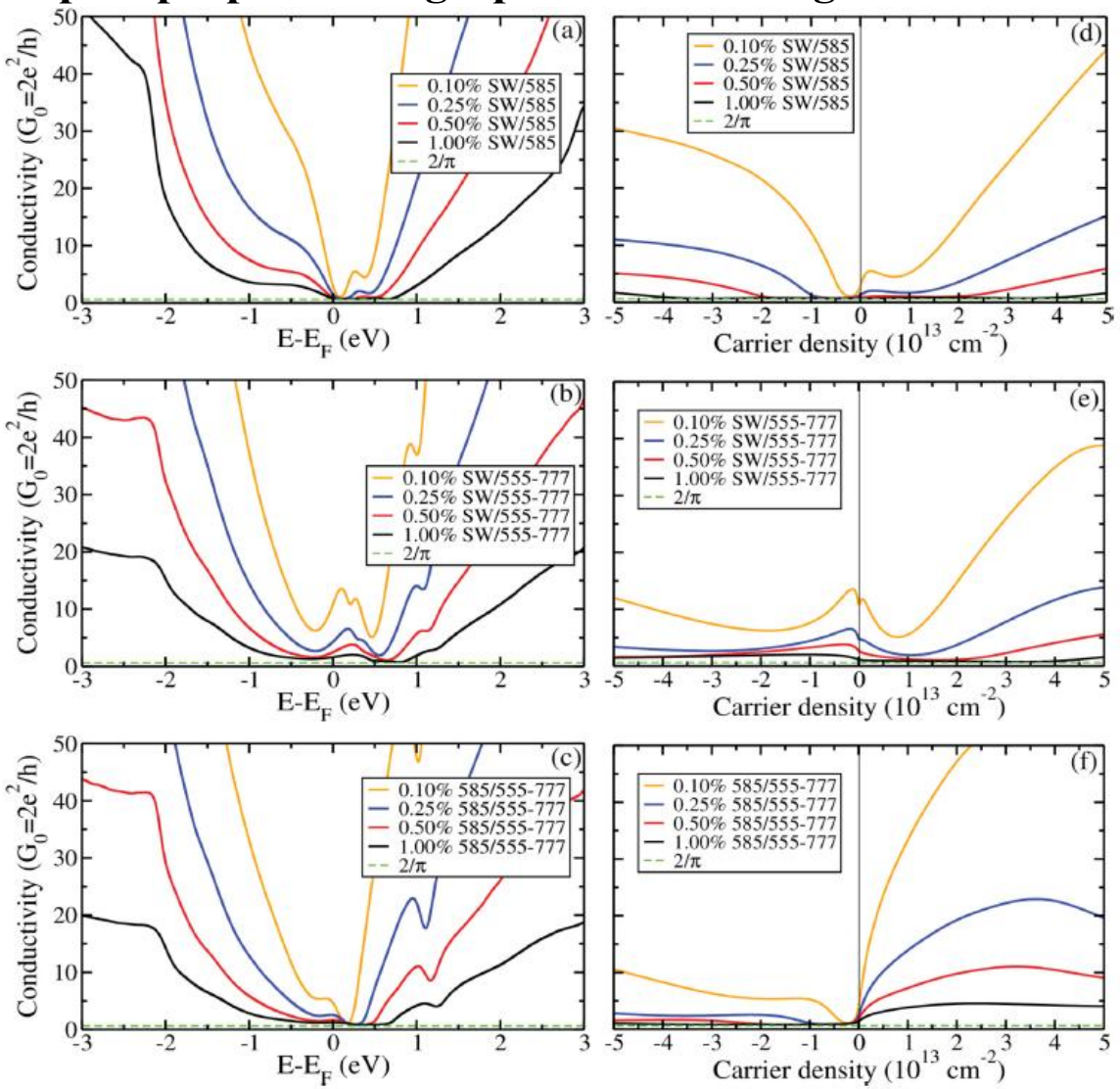


FIG. 12. (Color online) Conductivities as a function of charge carrier energy (left panels) and as a function of charge carrier density (right panels) of graphene planes with (*a*), (*d*) SW/585 defects, (*b*), (*e*) SW/555-777 defects, and (*c*), (*f*) 585/555-777 defects, for different defect concentrations ranging from *nd* = 0.1% to 1.0%

Aurélien Lherbier, Simon M.-M. Dubois, Xavier Declerck, *et al.*, Transport properties of graphene containing structural defects, *Phys. Rev. B* 86 075402 (2012)

III. Transport Anisotropy Induced by Strain or Defects in Graphene

Transport properties of graphene containing structural defects

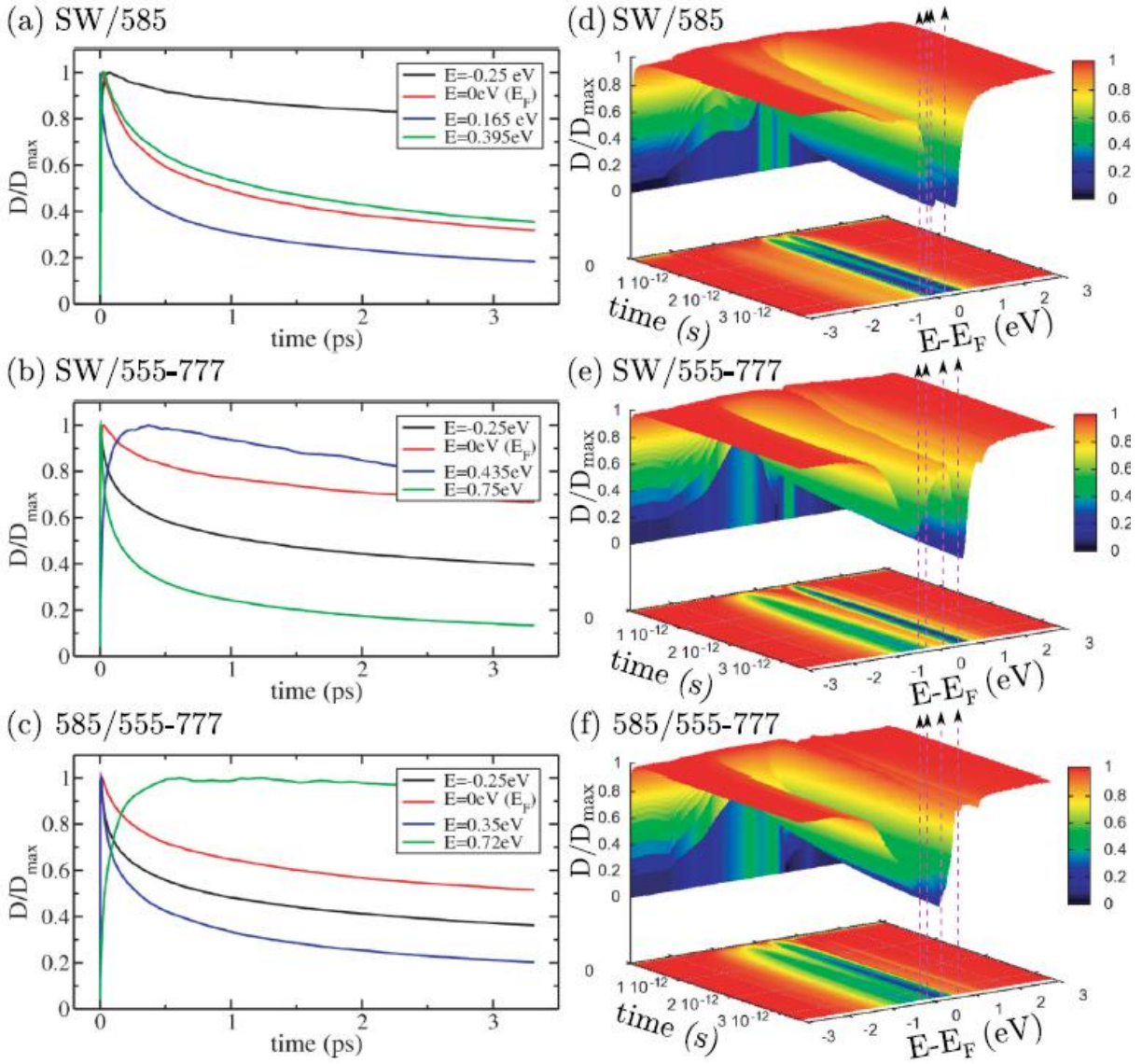


FIG. 13. (Color online) Normalized diffusivity as a function of time for four selected energies (left panels) and 3D plots as a function of time and energy (right panels) of graphene planes with (*a*), (*d*) SW/585 defects, (*b*), (*e*) SW/555-777 defects, and (*c*), (*f*) 585/555-777 defects, for a defect concentration *nd* = 1.0%

Aurélien Lherbier, Simon M.-M. Dubois, Xavier Declerck, *et al.*, Transport properties of graphene containing structural defects, *Phys. Rev. B* 86 075402 (2012)

III. Transport Anisotropy Induced by Strain or Defects in Graphene

In summary, the Kubo-Greenwood method allows one to quantify transport anisotropy caused by realistic perturbations in graphene. These insights are crucial for designing reliable graphene devices

This example also prepares perfectly for the next lecture, where the Kubo-Greenwood calculation in code using pristine graphene as the benchmark system will be implemented

IV. Kubo-Greenwood in Disordered and Strongly Correlated Systems

Disordered and strongly correlated materials are another field studied by Kubo formula. Here, the Kubo-Greenwood formula becomes particularly powerful because it can naturally incorporate scattering, localization, and interaction effects through the Green's function formalism

One of the most important applications of the Kubo-Greenwood method is in disordered systems. In the Anderson localization picture, strong disorder can localize electron wavefunctions, turning a metal into an insulator.

The Kubo-Greenwood conductivity remains finite only when states are extended. When we compute $\sigma(E)$ as a function of energy, we can identify the mobility edge — the critical energy where conductivity drops to zero. This is a direct computational signature of the localization transition

♦ Anderson localization: disorder-induced metal-insulator transition

♦ Mobility edge: energy separating extended and localized states

♦ How KG formula detects localization

IV. Kubo-Greenwood in Disordered and Strongly Correlated Systems

Anderson localization in Graphene

2. *The real-space Kubo-Greenwood method*

In the RSKG method [28–31], the zero-temperature dc electrical conductivity at energy E and correlation time τ can be expressed as

$$\sigma(E,\tau) = e^2 \rho(E) \frac{d\Delta X^2(E,\tau)}{2d\tau}, \tag{5}$$

where

$$\rho(E) = \frac{2\mathrm{Tr}\left[\delta(E-H)\right]}{\Omega} \tag{6}$$

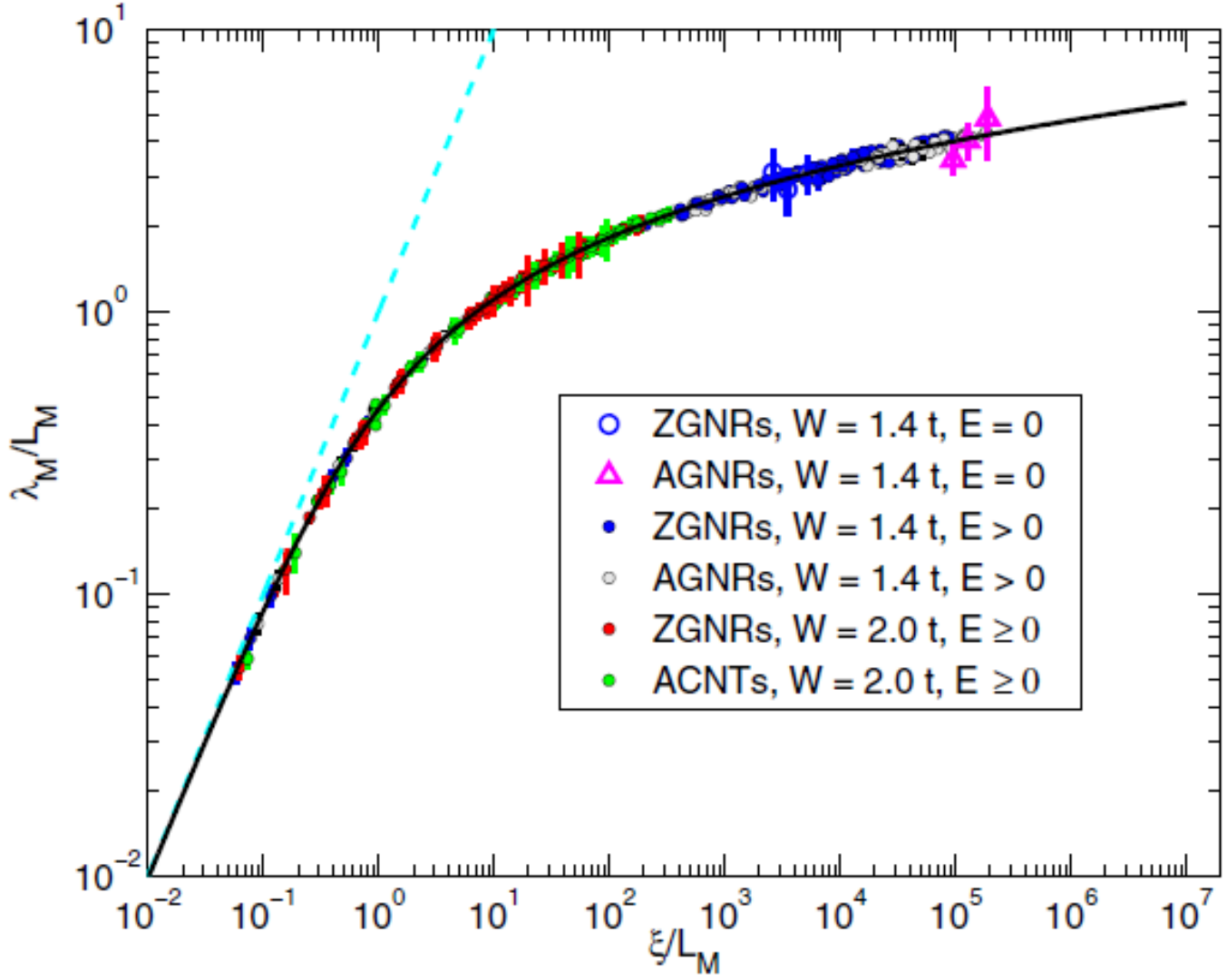
is the electronic density of states with the spin degeneracy taken into account. Note that the factors of 2 in the above two

Zheyong Fan, Andreas Uppstu and Ari Harju, Anderson localization in two-dimensional graphene with short-range disorder: One-parameter scaling and finite-size effects, *Phys. Rev. B* 89, 245422 (2014)

IV. Kubo-Greenwood in Disordered and Strongly Correlated Systems

Anderson localization in Graphene

FIG. 2. (Color online) One-parameter scaling of localization length. The localization length divided by the width, λ_M/L_M, is plotted as a function of ξ/L_M, where ξ is the 2D localization length obtained by fitting the data against the scaling curve. All the data from Fig. 1 with the last three largest M in each subfigure are considered. Abnormal data for the CNP in systems with weak disorder (W = 1.4 t) are emphasized. Due to the symmetry of the band structure, data with E < 0 from Fig. 1 are omitted. The solid line represents the scaling function given by Eq. (13) with $k = \pi$ and the dashed line represents the identity function $f(x) = x$. The error bars correspond to the error estimates of λ_M indicated in Fig. 1.



Zheyong Fan, Andreas Uppstu and Ari Harju, Anderson localization in two-dimensional graphene with short-range disorder: One-parameter scaling and finite-size effects, *Phys. Rev. B* 89, 245422 (2014)

IV. Kubo-Greenwood in Disordered and Strongly Correlated Systems

Anderson localization in Graphene

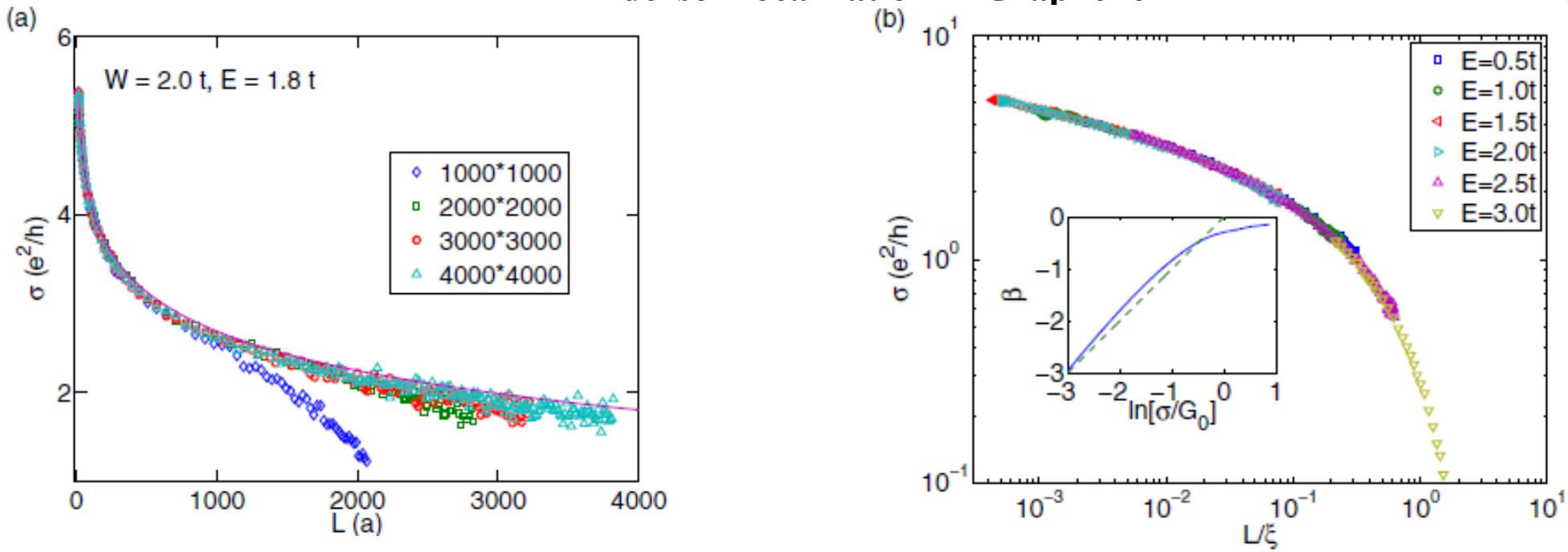


FIG. 5. (Color online) Conductivity for 2D graphene with W = 2.0 *t*. (*a*) Conductivity as a function of the propagating length of electrons for different simulation sizes $N_x * N_y$ (markers). The prediction from the weak-localization formula given by Eq. (20) is also shown (line). The energy considered here is E = 1.8 *t*. (b) Conductivity as a function of the reduced length L/ξ for a set of energy points. The 2*D* localization length ξ is taken to be the average over the results obtained shown in Fig. 4. The inset in (*b*) shows the renormalization group β function (solid line) calculated by using Eq. (21) after fitting σ as a smooth function of L/ξ. The dashed line in the inset represents $\beta = \ln(\sigma/G_0)$. Periodic boundary conditions are applied in both the transport and the transverse directions. The transport direction is taken to be along the zigzag edge; taking the transport direction to be along the armchair edge yields similar results.

Zheyong Fan, Andreas Uppstu and Ari Harju, Anderson localization in two-dimensional graphene with short-range disorder: One-parameter scaling and finite-size effects, *Phys. Rev. B* 89, 245422 (2014)

IV. Kubo-Greenwood in Disordered and Strongly Correlated Systems

Anderson localization in Graphene

♦ **Anderson localization in graphene with short-range disorder is fully described by one-parameter scaling theory, just as in conventional 2*D* disordered systems**

♦ **The proposed analytical scaling function with $k \approx \pi$ provides a reliable and efficient method to compute the 2D localization length from quasi-1D calculations**

♦ **Apparent anomalies at the CNP are finite-size effects, not failures of scaling theory — when the system width exceeds the mean free path by a sufficient margin, the CNP also follows the universal scaling**

♦ **The universal β function extracted from conductivity calculations confirms the absence of a metal-insulator transition in this system — all states are localized at zero temperature for sufficiently large systems**

Zheyong Fan, Andreas Uppstu and Ari Harju, Anderson localization in two-dimensional graphene with short-range disorder: One-parameter scaling and finite-size effects, *Phys. Rev. B* 89, 245422 (2014)

IV. Kubo-Greenwood in Disordered and Strongly Correlated Systems

Transport Property in the strongly correlated inhomogeneous multilayered nanostructures

The last step is to evaluate the dc conductivity matrix, which becomes

$$\sigma_{\alpha\beta}(0)=\frac{2e^2}{h}a^4t^2\int d\epsilon\,\rho^{2d}(\epsilon)\int d\omega\left(-\frac{df}{d\omega}\right)[\mathrm{Im}\,G_{\beta\alpha+1}(\epsilon,\omega)\mathrm{Im}\,G_{\alpha\beta+1}(\epsilon,\omega)+\mathrm{Im}\,G_{\beta+1\alpha}(\epsilon,\omega)\mathrm{Im}\,G_{\alpha+1\beta}(\epsilon,\omega)$$

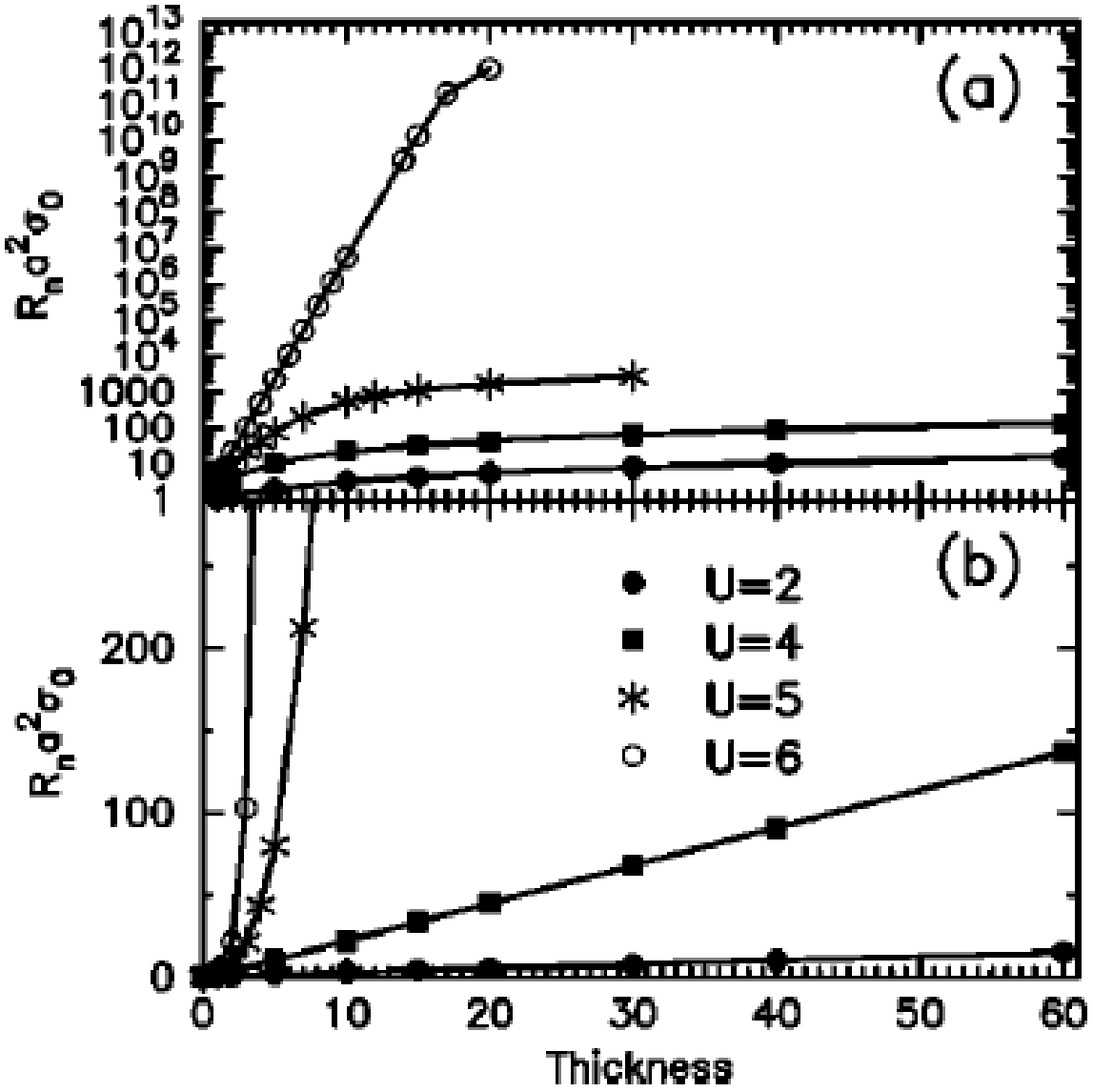
$$-\mathrm{Im}\,G_{\beta+1\alpha+1}(\epsilon,\omega)\mathrm{Im}\,G_{\alpha\beta}(\epsilon,\omega)+\mathrm{Im}\,G_{\beta\alpha}(\epsilon,\omega)\mathrm{Im}\,G_{\alpha+1\beta+1}(\epsilon,\omega)]. \quad (28)$$

J. K. Freericks, Dynamical mean-field theory for strongly correlated inhomogeneous multilayered nanostructures, Phys. Rev. B, 70, 195342 (2004)

IV. Kubo-Greenwood in Disordered and Strongly Correlated Systems

Transport Property in the strongly correlated inhomogeneous multilayered nanostructures

FIG. 11. Resistance-area product for nanostructures with U=2, 4, 5, and 6, and various thicknesses. Panel (a) is a semilogarithmic plot, while panel (b) is a linear plot. The temperature is T=0.01 in both panels. Note how the correlated insulator (U=6) has an exponential growth with thickness as expected for a tunneling process, but it turns over at the thickest junction, indicating a crossover to the incoherent transport regime. The U=5 data, which is close to the critical point for a MIT, has neither linear, nor exponential growth of its resistance-area product. The metallic cases (U=2 and 4) have perfect linear scaling of the resistance with current, with a nonzero intercept corresponding to the contact resistance. This may be surprising for U=4, because it is so strongly scattering (with a mean free path much less than a lattice spacing), that one would not think a semiclassical approach should apply there. The constant satisfies $\sigma_0 = 2e^2/ha^2$.



J. K. Freericks, Dynamical mean-field theory for strongly correlated inhomogeneous multilayered nanostructures, Phys. Rev. B, 70, 195342 (2004)

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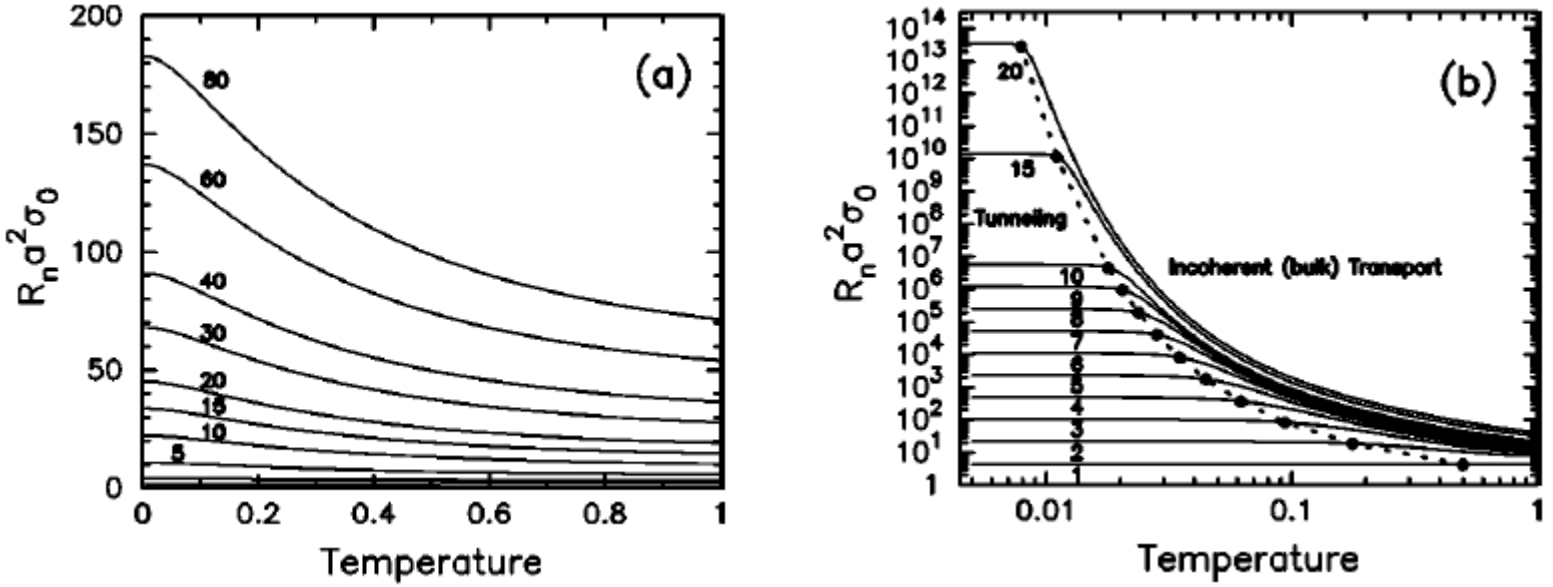


FIG. 12. Resistance-area product for nanostructures with (a) U = 4 and (b) U = 6 as a function of temperature [panel (a) is on a linear scale, and panel (b) is a log-log plot]. In panel (a) we include results for N=1, 2, (lowest two curves), 5, 10, 15, 20, 40, 60, and 80. Note how at each temperature there is a linear dependence of the resistance-area product with the thickness of the junction. Note further, that these junctions have anomalous temperature dependence for a metal (they actually look insulating in their dependence). In panel (b), we show the results for U = 6 with N = 1–10, 15 and 20. Note at low temperature we have tunneling, as the resistance-area product is weakly dependent on temperature, and the steps are equally spaced as a function of thickness, indicating exponential dependence on the thickness. At higher temperatures, there is a crossover to the incoherent transport regime, with the resistance-area product picking up a strong T dependence, and scaling linearly with the thickness. The dotted line that connects the solid dots is a plot of the resistance-area value at the temperature where E_{Th} = T which determines the crossover.

J. K. Freericks, Dynamical mean-field theory for strongly correlated inhomogeneous multilayered nanostructures, Phys. Rev. B, 70, 195342 (2004)



That's all for this session

Thank you for your attention and great participation